

PUNJAB DRAINAGE CHANNELS

Sector: Agriculture

Hazard addressed: Flood

Assets covered: Rice and Cotton



Description: This measure involves construction and rehabilitation of drainage channels to safely convey excess floodwater and storm runoff away from vulnerable agricultural and settlement areas. The intervention reduces flood inundation, minimizes waterlogging, improves drainage efficiency, and protects crops, infrastructure, and livelihoods from flood-related damages.

Priority Districts: Sialkot, Jhang, Toba Tek Singh, Khanewal, Vehari, Bahawalnagar, Bahawalpur, Lodhran, Rajanpur, DG Khan, Muzaffargarh, Multan, Rahim Yar Khan

TECHNICAL SPECIFICATIONS/DIMENSION



Unit of implementation: Kilometer (km) of drainage channel

Scale assumptions: Applicable in flood-prone agricultural districts and low-lying areas vulnerable to drainage congestion and seasonal flooding.

Key design parameters: Drainage channel width and depth, hydraulic flow capacity, embankment stabilization, sediment management, culverts, and outfall structures designed according to local hydrological conditions.

Time horizon: Effective from 2041 till 2050.

IMPACT



Risk reduction level: High

Time to effectiveness: Medium-term following completion of channel construction and operationalization.

Expected impact: Reduced flood inundation and waterlogging, improved drainage efficiency, protection of crops and infrastructure, and enhanced resilience of agricultural systems against flood hazards.

BENEFICIARIES



Primary beneficiaries: Farmers, irrigation departments, local governments, rural communities, and disaster management authorities.

Distributional aspects: Particularly beneficial for vulnerable agricultural communities located in flood-prone and poorly drained areas.

COST ASSUMPTIONS



Unit cost: Drainage channels: PKR 25.97 million per km (approximately USD 92,750 per km).

Initial Investment Cost (CAPEX): Drainage channels: PKR 25.23 million per km (approximately USD 90,107 per km). Includes excavation, embankment construction, culverts, drainage structures, and associated civil works.

Annual O&M costs (OPEX): Drainage channels: PKR 0.7569 million per km/year (approximately USD 2,703 per km/year). Includes desilting, vegetation removal, structural repairs, and routine maintenance.

CO-BENEFITS & TRADE-OFFS



Environmental: Reduced standing water and waterlogging, improved soil conditions, and reduced erosion risks.

Social: Improved safety and livelihood protection for flood-prone rural communities.

Economic: Reduced crop and infrastructure losses due to flooding and improved agricultural productivity.

Potential risks: High capital investment requirements, sediment accumulation, encroachment issues, and maintenance challenges during extreme flood events.

IMPLEMENTATION CONSIDERATIONS



Enabling conditions: Hydrological assessments, engineering design expertise, institutional coordination, financing support, and land availability.

Required institutions: Punjab Irrigation Department, Agriculture Department Punjab, PDMA Punjab, Local Government Department, and development partners.

Barriers: Land acquisition constraints, sedimentation, drainage blockage, weak maintenance systems, and limited long-term financing.